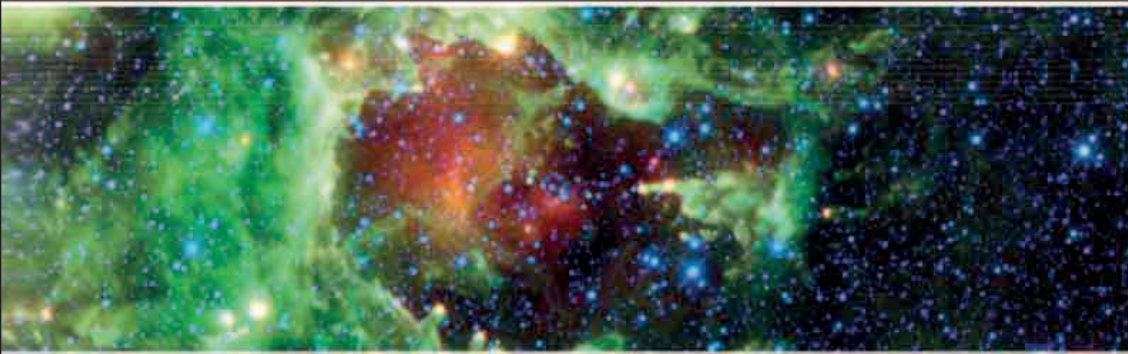




Immanuel Kant believed, as do many scientists today, that the Solar System originated as a cloud of dust between the stars.

In Russia, polymath **Mikhail Lomonosov** proved the law of **conservation of matter** by showing that when lead plates are heated inside a jar the collective weight of jar and contents stays the same, although the materials have altered. His findings predated by nearly 30 years the similar law formulated by French chemist Antoine Lavoisier (see 1781–82).

German philosopher **Immanuel Kant** (1724–1804) developed the idea of the nebular hypothesis, first suggested by Swedish thinker and visionary **Emanuel Swedenborg** (1688–1772) in the 1730s. This theory postulates that the Solar System originated in a rotating gas cloud

MIKHAIL LOMONOSOV (1711–65)

Little known in the West, Lomonosov was born a peasant and went on to become a pioneer in physics, chemistry, and astronomy, a poet, and a key thinker in the Russian Enlightenment. His achievements include the discovery of planet Venus's atmosphere, and devising theories for light waves and iceberg formation.



that collapsed inward under its own gravity, the matter forming into the Sun and planets.

Destruction of a city
Over 30,000 people were killed by the massive earthquake that rocked the Portuguese city of Lisbon in November 1755.



“A PARTICLE... OF POINTS QUITE HOMOGENEOUS, **SUBJECT TO A LAW OF FORCES...** MAY EITHER **ATTRACT, REPEL, OR HAVE NO EFFECT...** ON ANOTHER PARTICLE...”

Ruder Bošković, Croatian scientist, in *Theory of Natural Philosophy*, 1758

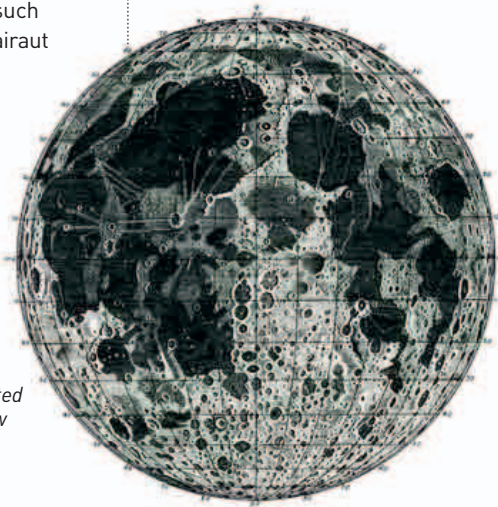
THE IDEA THAT THINGS ARE MADE FROM ATOMS

had been important since the early 17th century. But **Ruder Bošković**, living in Venice at the time, went further and developed his own atomic theory, which he explained in his book *Theoria Philosophiae* (*Theory of Natural Philosophy*). Bošković suggested that matter is built from pointlike particles interacting in pairs.

In France, astronomer **Alexis Claude Clairaut** (1713–65), who had earlier made a name for himself with his theory on why Earth must be flattened at the poles, developed a **theory about comets**. He suggested that Halley's comet, due to reappear in 1759, might be subject to unknown gravitational forces, such as another comet. Clairaut also compiled lunar tables, but they were not as important or as accurate as those made in Göttingen, in Germany, by astronomer **Tobias**

Mayer (1723–62) and used to calculate longitude at sea. Also working on longitude calculations at about this time was English clockmaker **John Harrison** (1693–1776), whose H3 chronometer proved precise enough to be used for longitude calculations under all conditions.

In Switzerland, the three engineering **Grubenmann brothers**—Jakob (1694–1758), Johannes (1707–71), and Hans (1709–83)—erected the world's longest road bridges, including a 220 ft- (67 m-) long bridge at Reichenau over a tributary of the Rhine.



Moon map

German astronomer Tobias Mayer's close study of the Moon resulted in the first maps to show accurately the positions of the craters on the lunar surface.

1756 Russian polymath Mikhail Lomonosov proves the law of conservation of matter

1756 British engineer John Smeaton develops water-setting concrete

1757 English clockmaker John Harrison completes his H3 chronometer

1757 German astronomer Tobias Mayer presents accurate tables of the Moon's motion

1758 French astronomer Alexis Claude Clairaut suggests Halley's Comet may be steered by another comet

1758 Croatian scientist Ruder Bošković publishes his atomic theory

1758 Swiss engineers the Grubenmann brothers erect the world's longest road bridges